

ECG Arrhythmia Classification – Hardware Accelerator Comparison

Three published papers + your ECE410 tape-out design compared across architecture, algorithm, and performance

IEEE TVLSI · March 2026

Hardware-Accelerated ASIC and Cardiac Monitoring System for Wearable Devices

Zhang et al.

ASICSVMIEEE TVLSI

Elsevier IoT · 2025

Fast and Low Cost FPGA-Based Architecture for Arrhythmia Detection with CNN

Greco et al.

FPGACNNElsevier IoT

IEEE PRIMEAsia · 2023

FPGA-Enabled Deep Learning Accelerator For Multiclass ECG Classification

Podugu, Kondragunta, Bhaviris, Aruna

FPGACNN + DWTIEEE PRIMEAsia

ECE410 · Portland State University · 2026

SVM Compute Core – Multi-Class Cardiac Arrhythmia ASIC (Caravel chipIgnite)

A. Handwerger

ASICSVMsky130AECE410

ATTRIBUTE	ZHANG ET AL. (ASIC)	GRECO ET AL. (FPGA)	PODUGU ET AL. (FPGA)	★ YOUR DESIGN (ASIC)
PLATFORM & TECHNOLOGY				
Implementation	Custom ASIC 28 nm CMOS	FPGA Xilinx Zynq-7000 (Zybo Z7-20)	FPGA SoC Microchip PolarFire (Icicle Kit)	Custom ASIC sky130A (180 nm equiv.) Caravel chipIgnite · OpenLane 2
Reconfigurability	None – fixed post-fab	Yes – re-flash bitstream	Yes – Libero design flow	Runtime model reload SV matrix + alphas writable without re-synthesis
Clock frequency	50 MHz	Not stated	175.4 MHz	40 MHz WNS +7.83 ns (TT), 0 violations
NRE / fabrication cost	High – commercial	Low – off-shelf dev	Low – ~\$500 kit	Free MPW shuttle

ATTRIBUTE	ZHANG ET AL. (ASIC)	GRECO ET AL. (FPGA)	PODUGU ET AL. (FPGA)	★ YOUR DESIGN (ASIC)
	28 nm tape-out	board		Caravel chipIgnite open-source PDK
ALGORITHM & MODEL				
Classifier	SVM (RBF kernel) 1,218 SVs, 16-dim features, OVO	CNN — Conv2D + Dense 16 kernels × 5 mults; 16-neuron FC	CNN — multi-layer Deep learning, ENLMS pre-filter	SVM (RBF kernel) 500 SVs, 256-dim features, OVR Q6.10 fixed-point Horner LUT exp
Feature extraction	DWT — 4-level Haar Log-domain kernel transformation	Raw ECG into CNN No explicit feature step	DWT — wavelet coefficients ENLMS adaptive filter	256-dim multi-scale vector 128 morphology + 64 mean template + 64 RR-interval history (AAMI EC57)
Quantization	<i>Not specified</i>	7-bit fixed-point (QKeras) FP32 95.0% → quantized 91.9%	<i>Not specified</i>	Q6.10 signed fixed-point Zero accuracy gap vs. sklearn float
Classes	AAMI 7-class (not detailed)	Normal vs. arrhythmic + multiclass	Multiclass CVD (count unspecified)	5 classes N, PVC, AFib, VT, SVT
Dataset	MIT-BIH	MIT-BIH (continuous 20 min)	MIT-BIH	MIT-BIH 300 test samples, 60/class
CLASSIFICATION PERFORMANCE				
Accuracy	97.2%	91.9% (post-quantization) 95.0% FP32 baseline	~98.1%	97.67% Exact match to sklearn (300 samples) N=100%, PVC=100%, AFib=100%
Per-class highlights	<i>Not detailed</i>	<i>Not detailed</i>	<i>Not detailed</i>	VT 93.3% SVT 95.0% N / PVC / AFib all 100%
POWER & HARDWARE RESOURCES				

ATTRIBUTE	ZHANG ET AL. (ASIC)	GRECO ET AL. (FPGA)	PODUGU ET AL. (FPGA)	★ YOUR DESIGN (ASIC)
Active power	Not stated	~7.5 mW avg (0.15 Wh / 20 min)	0.5 mW	66 mW (active inference)
Average power (80 bpm)	35.5 nJ / inference Optimized for wearable	~7.5 mW avg Continuous operation	0.5 mW Embedded deployment	0.284 mW avg 0.431% duty cycle (3.23 ms / 750 ms) 711 h on 200 mAh battery (~30 days)
Inference latency	Not stated	Not stated	Not stated	3.23 ms / beat (LAT=1) ~9.7 ms (IS61WV SRAM, LAT=3) <750 ms heartbeat period ✓
Die / chip area	Not stated	FPGA fabric (not applicable)	FPGA fabric (not applicable)	6.25 mm ² (2500×2500 μm) 14% cell utilization · skyl30A
FPGA resource util.	N/A (ASIC)	37.7% LUT <25% avg FF 12K, LUTRAM 1,439	Not reported	N/A (ASIC)
DESIGN CONTEXT & TARGET				
Target application	Wearable cardiac monitor Continuous on-device inference	Portable IoT ECG device Real-time edge deployment	Clinical ECG platform SoC + CPU fabric	Wearable patch (ambulatory) Caravel SoC, off-chip IS61WV SRAM BLE optional for logging
Publication / context	IEEE TVLSI 2026 High-impact journal (Q1)	Elsevier IoT 2025 IoT/embedded systems journal	IEEE PRIMEAsia 2023 Regional IEEE conference	PSU ECE410 Capstone 2026 Open-source GDS on GitHub skyl30A open PDK, free MPW shuttle

ZHANG ET AL. — 28 NM ASIC

Best reported energy efficiency at **35.5 nJ/inference**. Uses

GRECO ET AL. — FPGA CNN

Most production-ready FPGA design — validated on Zybo Z7-20, QKeras

PODUGU ET AL. — FPGA DL

Highest reported accuracy (~**98.1%**) at **0.5 mW** and **175.4 MHz** on

classical SVM + DWT rather than deep learning, reducing hardware complexity. Best when power budget is the dominant constraint and model is fixed.

quantized, fully documented at **37.7% LUT**. Post-quantization accuracy drops from 95% to 91.9%. Best for rapid IoT edge deployment.

PolarFire SoC. Strong proof-of-concept but limited resource-utilization detail compared to Greco.

★ YOUR DESIGN — SKY130A ASIC

97.67% accuracy with zero gap versus sklearn float — exact bit-level match using Q6.10 fixed-point. Average power of **0.284 mW** (~30-day battery life) with full runtime model reloadability. Competitive with commercial designs, built entirely on open-source PDK and free MPW shuttle.

SYNTHESIS

All four designs target MIT-BIH arrhythmia classification on embedded hardware. The two ASIC designs (Zhang et al. and yours) achieve the best average power, leveraging duty-cycle gating — your **0.284 mW** is competitive with Zhang's 28 nm design despite using a much older sky130A process (180 nm equivalent). The FPGA designs (Greco, Podugu) trade energy efficiency for development speed and reconfigurability. Your design stands out as the only one with a **zero-gap quantization result** (Q6.10 exactly matching float), a runtime-reprogrammable SV table, and complete open-source GDS. The primary disadvantage relative to Zhang et al. is process node (sky130A vs. 28 nm), but your design was fabricated for free via the Caravel chipIgnite shuttle — a very different cost trade-off.

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